

SIMATS ENGINEERING

Department of [AI & DS]

CASE STUDY ANALYSIS

AI-Powered Smart Volunteer and NGO Resource Management Platform

Student Name: R. Indhu

Register Number: 192524332

Department: AI & DS

Course Name: CSA1017-Software Engineering

Faculty Name: R. Latha

Academic Year: 2025-2029

*Submitted in partial fulfilment of the requirements of
CSA1017 — Software Engineering*

TABLE OF CONTENTS

TABLE OF CONTENTS	2
1. Understand the Problem Statement	3
1.1 Problem Description	3
1.2 Objectives of the Proposed System.....	3
1.3 Scope of the Problem	4
2. Key Stakeholders and Challenges	4
2.1 Identification of Stakeholders	4
2.2 Roles and Responsibilities	4
2.3 Stakeholder Challenges and Issues	5
3. Current System Analysis	6
3.1 Existing System / Current Process	6
3.2 Strengths of Existing System	6
3.3 Limitations of Existing System	7
3.4 Problems and Gaps	7
4. Proposed Software Solution	7
4.1 Proposed Solution	7
4.2 Major Functional Modules.....	8
4.3 Functional Requirements	8
4.4 Non-Functional Requirements.....	9
4.5 Application of Software Engineering Principles.....	9
5. SDLC / Framework Justification	11
5.1 Selected Methodology	11
5.2 Description of the Methodology.....	11
5.3 Reasons for Selecting It	11
5.4 How It Supports Project Development	12
5.5 How It Supports Testing and Maintenance.....	Error! Bookmark not defined.
6. Expected Outcomes and Limitations	12
6.1 Expected Outcomes.....	12
6.2 Expected Benefits	12
6.3 How the Solution Addresses the Challenges.....	13
6.4 Potential Risks	13
7. Conclusion	14
8. References	14

1. Understand the Problem Statement

1.1 Problem Description

Non-Governmental Organizations (NGOs) and volunteer groups play a critical role in delivering humanitarian services, education, healthcare, disaster relief, and community development programs. The effectiveness of these initiatives depends heavily on how efficiently volunteers are recruited, matched to appropriate activities, and monitored throughout project execution, as well as on how transparently donations and resources are tracked. In practice, most NGOs, particularly small and medium-sized organizations, continue to rely on fragmented, manual, and paper- or spreadsheet-based systems to manage these functions. Volunteer registration is often handled through phone calls or physical forms, task assignment depends on the personal judgement and memory of coordinators rather than on a structured matching process, and donation records are maintained separately from project and volunteer data.

1.2 Objectives of the Proposed System

The proposed platform is guided by the following core objectives, derived directly from the operational needs of NGOs and volunteer organizations:

1. Manage volunteer registrations digitally, replacing manual, paper-based, and spreadsheet-driven enrolment processes.
2. Match volunteers to suitable activities using Artificial Intelligence, based on skills, interests, availability, and location.
3. Monitor the progress of NGO projects in real time, enabling timely intervention when milestones are at risk.
4. Track donations and resources transparently, from receipt to utilization, for every project.
5. Optimize the allocation of volunteers and resources across concurrent projects to maximize social impact.
6. Generate organizational reports automatically for internal review, donor communication, and regulatory compliance.

7. Improve communication among stakeholders, including volunteers, coordinators, donors, and beneficiaries.
8. Enhance operational transparency across all NGO activities, strengthening donor and community trust.

1.3 Scope of the Problem

The platform is intended for use by four primary user groups: volunteers seeking meaningful engagement opportunities, NGO administrators and project coordinators responsible for planning and oversight, donors who contribute funds or resources and wish to track their utilization, and system administrators who maintain the platform. The solution is designed as a cloud-hosted system accessible through a web-based administrative dashboard and a mobile application for volunteers and donors, incorporating GIS-based location services for proximity-based volunteer matching. Matters outside the scope of this analysis include the detailed accounting/taxation compliance mechanisms of individual NGOs, the specific legal frameworks governing NGO registration in different jurisdictions, and the underlying payment-gateway certification processes, which are treated as external, integrated services rather than components to be designed in depth.

2. Key Stakeholders and Challenges

2.1 Identification of Stakeholders

The proposed platform serves a diverse set of stakeholders, each with distinct needs and expectations. The primary stakeholders identified for this case study are volunteers, NGO administrators and coordinators, donors, beneficiaries or the communities served, project managers, IT/system administrators responsible for platform upkeep, and, indirectly, government or regulatory bodies that oversee NGO compliance and reporting.

2.2 Roles and Responsibilities

Table 2.1 summarizes the role, core responsibilities, and principal challenges faced by each stakeholder group under the current, largely manual mode of operation.

Stakeholder	Role	Responsibility	Challenges
Volunteers	Service providers / field participants	Register on the platform, accept and complete assigned tasks, update task status, provide feedback	Difficulty discovering suitable opportunities; limited visibility into the impact of their contribution; inconsistent communication from coordinators
NGO Administrators / Coordinators	Programme managers	Create projects, assign volunteers to tasks, allocate resources, monitor progress, communicate with stakeholders	Manual assignment is time-consuming and error-prone; fragmented records across spreadsheets and paper registers; delayed, reactive decision-making
Donors	Resource / fund contributors	Contribute funds or material resources, review utilization reports, renew or extend support	Lack of transparency into how donations are used; no real-time visibility of project outcomes funded by their contribution
Beneficiaries / Communities	Recipients of NGO services	Communicate needs, participate in programmes, provide feedback on service quality	Delayed or inconsistent service delivery caused by poor coordination between volunteers and NGO staff
Project Managers	Oversight of specific initiatives	Track milestones, manage timelines, allocate volunteers and resources within a project	No unified dashboard for real-time status; reliance on verbal or informal progress updates
IT / System Administrators	Technical support and platform maintenance	Maintain system uptime, manage data security and backups, support integrations	Supporting multiple disconnected legacy tools; difficulty enforcing consistent data-security practices
Government / Regulatory Bodies	Compliance and oversight authority	Review NGO activity reports, verify fund utilization, ensure regulatory compliance	Absence of standardized, auditable reporting formats from NGOs using manual systems

Table 2.1: Stakeholders, Roles, Responsibilities, and Challenges

2.3 Stakeholder Challenges and Issues

Across all stakeholder groups, a recurring theme emerges: the absence of a shared, real-time source of information. Volunteers cannot easily discover tasks that match their skills and availability, which lowers engagement and increases attrition. Coordinators must manually cross-reference volunteer lists, task requirements, and project calendars, a process that does not scale as the number of active projects grows. Donors, lacking a transparent tracking mechanism, often base continued support on trust rather than verifiable evidence of impact, which can erode confidence over time. Beneficiaries experience inconsistent service delivery

whenever volunteer availability and task assignment are not properly synchronized. These interconnected challenges collectively motivate the need for an integrated, AI-powered platform that gives every stakeholder timely, accurate, and role-appropriate visibility into NGO operations.

3. Current System Analysis

3.1 Existing System / Current Process

In the current, largely manual environment, volunteer registration is typically carried out through phone calls, in-person visits, or paper forms that are later transcribed into spreadsheets by NGO staff. Task assignment relies on the coordinator's personal knowledge of available volunteers and their perceived suitability, rather than on a structured evaluation of skills, interests, availability, or proximity to the activity location. Once assigned, volunteers are notified informally through phone calls or messaging applications, and their progress is tracked through periodic verbal check-ins rather than a documented workflow.

At the end of a reporting period, administrators must manually compile data from these disconnected sources to produce progress reports for management, donors, or regulatory authorities. Figure 3.1 illustrates this current workflow and highlights the disconnected nature of the process.

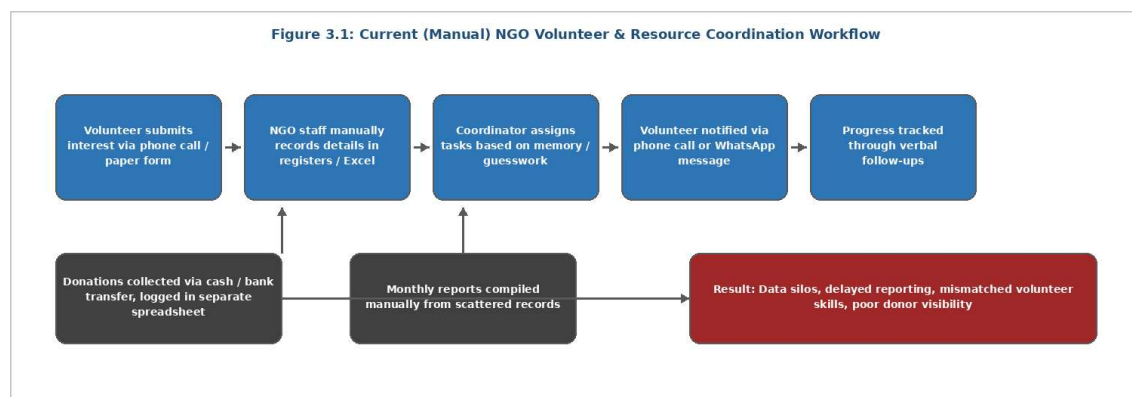


Figure 3.1: Current (Manual) NGO Volunteer and Resource Coordination Workflow

3.2 Strengths of Existing System

Despite its limitations, the existing manual process offers a few practical advantages, particularly for very small NGOs operating with minimal budgets and infrastructure. It requires no upfront technology investment, is familiar to long-serving staff, and remains operable even

without reliable internet access. Table 3.1 presents a structured comparison of these strengths against the corresponding limitations of the current system.

3.3 Limitations of Existing System

Strengths of Existing System	Limitations of Existing System
Low barrier to entry; no technical training or infrastructure investment required	No centralized database; volunteer, task, and donation data remain in disconnected files
Familiar and trusted by long-serving staff and community volunteers	High dependency on individual staff memory and informal communication channels
Flexible and easily adapted to ad-hoc, unplanned activities	No real-time visibility into project status for administrators or donors
Low direct monetary cost for very small NGOs with limited budgets	Volunteer-task matching is subjective and does not consider skills or location systematically
Personal relationships between coordinators and volunteers can motivate engagement	Donation records are prone to loss, duplication, or manual entry errors
No dependency on internet connectivity for basic day-to-day coordination	Reporting is slow, labour-intensive, and difficult to standardize across projects

Table 3.1: Strengths versus Limitations of the Existing (Manual) System

3.4 Problems and Gaps

The analysis of the current system reveals several critical gaps that directly justify the need for the proposed platform. First, there is no centralized, real-time source of truth for volunteer, task, project, or donation data, which forces stakeholders to rely on fragmented and often outdated information. Second, the absence of an intelligent matching mechanism means volunteers are frequently assigned to tasks that do not align well with their skills, interests, or availability, reducing both efficiency and volunteer satisfaction.

4. Proposed Software Solution

4.1 Proposed Solution

The proposed solution is an AI-Powered Smart Volunteer and NGO Resource Management Platform: a cloud-hosted system accessible through a web-based administrative dashboard for NGO staff and a mobile application for volunteers and donors. The platform digitizes volunteer registration, applies an AI-driven recommendation engine to match volunteers with suitable tasks based on skills, interests, availability, and geographic proximity (using GIS-based

location data), monitors the progress of NGO projects in real time, and tracks donations and resources from receipt through to utilization

4.2 Major Functional Modules

The platform is organized into six major functional modules, illustrated in Figure 4.2, each responsible for a distinct area of functionality while remaining fully integrated with the others through a common data layer:

- Volunteer Registration and Profile Module — manages volunteer sign-up, skills and interest capture, availability scheduling, and profile verification.
- AI Volunteer-Task Matching Module — applies machine learning models to recommend the best-fit volunteers for a given task or project based on skills, past performance, location, and availability.
- Task and Project Monitoring Module — enables coordinators to create projects and tasks, assign volunteers, and track milestone completion in real time.
- Donation and Resource Tracking Module — records monetary and in-kind donations, links each contribution to specific projects, and tracks resource utilization and inventory.
- Reporting and Analytics Dashboard Module — generates real-time dashboards and periodic reports for administrators, donors, and regulatory submissions.

4.3 Functional Requirements

9. The system shall allow volunteers to register, create profiles, and specify skills, interests, and availability.
10. The system shall use an AI-based algorithm to recommend suitable volunteers for each task or project.
11. The system shall allow coordinators to create, assign, and track tasks and projects, with milestone status updates.
12. The system shall record donations (monetary and in-kind), generate receipts, and link contributions to specific projects.

13. The system shall provide real-time dashboards displaying volunteer engagement, project progress, and resource utilization.

4.4 Non-Functional Requirements

- Scalability — the platform must support a growing number of NGOs, volunteers, and concurrent projects without performance degradation.
- Security — volunteer, donor, and financial data must be protected through encryption, authentication, and role-based access control.
- Usability — the interface must be intuitive for users with varying levels of digital literacy, including volunteers and community-level NGO staff.
- Reliability — the system must maintain high availability, particularly during time-critical disaster-relief coordination.
- Performance — AI-based matching and dashboard queries must return results within acceptable response times, even under peak load.
- Maintainability — the modular architecture must allow individual modules to be updated or extended without disrupting the entire system.
- Interoperability — the platform must integrate with third-party services such as GIS/maps APIs, payment gateways, and SMS/email gateways.

4.5 Application of Software Engineering Principles

Modularity

The platform is decomposed into six independent but interoperable modules (registration, matching, monitoring, donation tracking, reporting, and communication), each exposed through well-defined APIs.

Scalability

The system is designed as a set of cloud-hosted microservices behind an API gateway, allowing individual services — such as the AI matching engine during high-demand disaster-relief periods — to be scaled independently based on load, without requiring the entire platform to be scaled uniformly.

Maintainability

Clear separation of concerns between the presentation, application/service, and data layers (as shown in Figure 4.1) ensures that changes to one layer, such as updating the AI matching model, have minimal impact on other layers, reducing the cost and risk of ongoing maintenance.

Reliability

Cloud infrastructure with automated backup, monitoring, and failover mechanisms ensures the platform remains available and resilient, which is particularly important during time-sensitive disaster-relief and healthcare coordination activities.

Usability

The mobile application for volunteers and donors and the web dashboard for administrators are designed around simple, task-oriented workflows, minimizing the number of steps required to register, accept a task, or make a donation, so that users with limited digital literacy can operate the system confidently.

Security

Role-based access control, encrypted data storage and transmission, and secure integration with payment gateways protect sensitive volunteer, beneficiary, and financial information, helping the platform comply with data-protection expectations relevant to NGO operations.

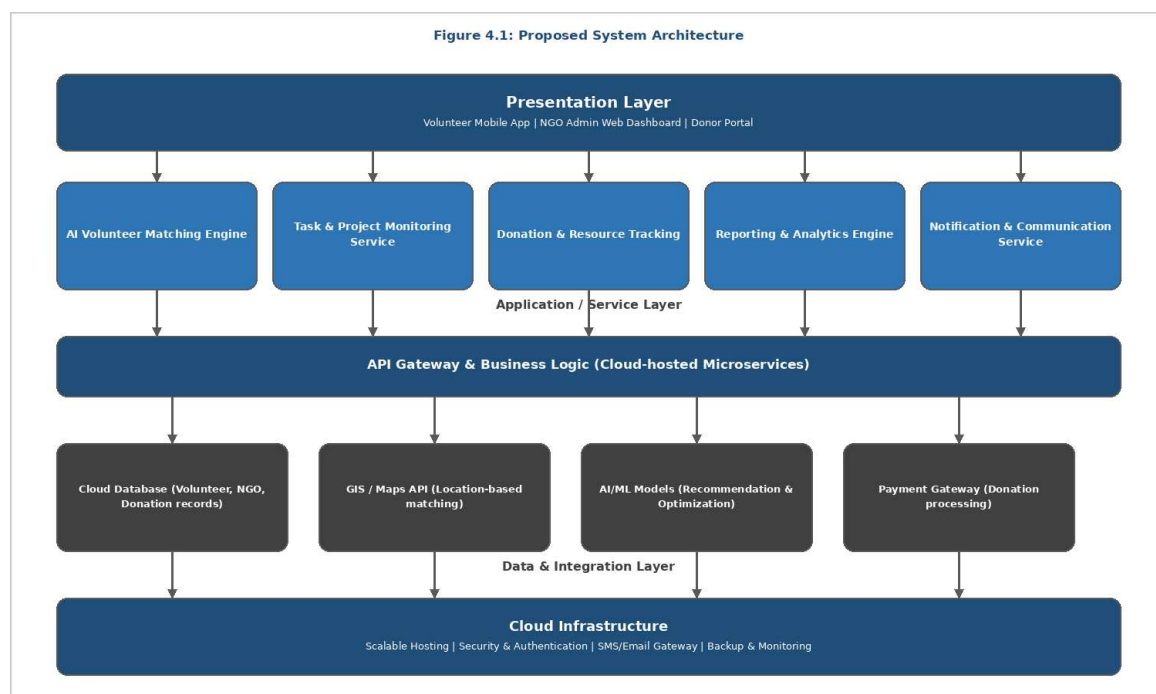


Figure 4.1: Proposed System Architecture

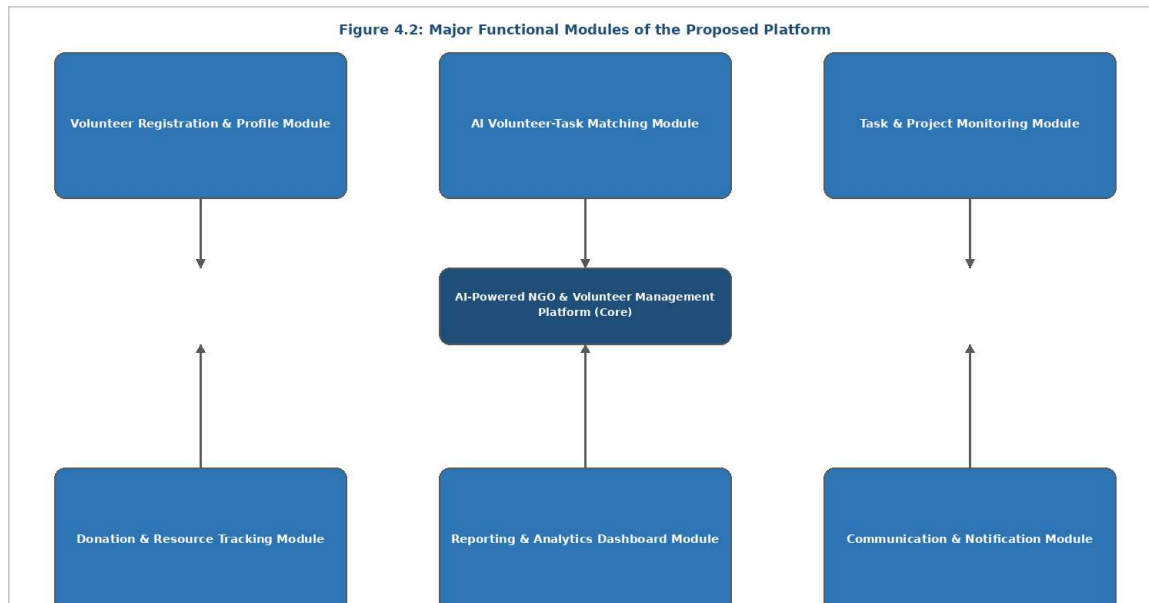


Figure 4.2: Major Functional Modules of the Proposed Platform

5. SDLC / Framework Justification

5.1 Selected Methodology

The Agile methodology, implemented through the Scrum framework, is selected as the most appropriate software development approach for the AI-Powered Smart Volunteer and NGO Resource Management Platform.

5.2 Description of the Methodology

Agile is an iterative and incremental software development approach that delivers working software in short cycles called sprints, typically two to four weeks in duration. Scrum, a widely used Agile framework, organizes work into a prioritized product backlog, sprint planning sessions, daily stand-up meetings, sprint reviews, and retrospectives, with a Product Owner representing stakeholder interests and a Scrum Master facilitating the process.

5.3 Reasons for Selecting It

Agile/Scrum is particularly well suited to this project for several reasons directly tied to the nature of the problem. NGO operational requirements frequently evolve as new project types, donor requirements, or regulatory reporting formats emerge, and Agile's iterative structure

accommodates such evolving requirements far more effectively than a rigid, sequential model. The platform also involves several distinct but interdependent modules — volunteer management, AI matching, donation tracking, and reporting — that can be developed and delivered incrementally, allowing NGOs to begin benefiting from core functionality (such as digital volunteer registration) before more advanced features (such as AI-based matching) are fully refined.

5.4 How It Supports Project Development

Under Scrum, the product backlog would be organized around the platform's major modules, with early sprints focused on core volunteer registration and task management functionality, followed by sprints dedicated to the AI matching engine, donation tracking, and reporting dashboards. Regular sprint reviews allow NGO stakeholders to evaluate working increments of the system and provide feedback that directly shapes subsequent sprints, reducing the risk of building features that do not match real-world NGO workflows.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

- Digitally managed, centralized volunteer registration replacing manual and paper-based processes.
- AI-driven matching of volunteers to suitable activities, improving assignment accuracy.
- Real-time monitoring of NGO project progress across all active initiatives.
- Transparent, end-to-end tracking of donations and resource utilization.
- Optimized allocation of volunteers and resources across concurrent projects.
- Automated generation of organizational and compliance reports.

6.2 Expected Benefits

- Improved volunteer participation and retention through better-matched, more engaging assignments.
- Efficient resource allocation resulting in reduced duplication of effort and wasted capacity.

- Better project management through real-time visibility into milestones and risks.
- Enhanced organizational transparency, strengthening accountability to donors and regulators.
- Increased donor confidence due to verifiable, real-time tracking of contributions.
- Reduced administrative workload for NGO coordinators, freeing time for programme delivery.

6.3 How the Solution Addresses the Challenges

The AI matching module directly addresses the mismatch between volunteer skills and assigned tasks identified in Section 2.3, while the centralized project-monitoring module eliminates the fragmented, memory-dependent coordination described in Section 3.1. The donation-tracking module resolves the donor-transparency gap by linking every contribution to a specific, trackable project outcome, and the reporting and analytics dashboard removes the delay associated with manually compiled, retrospective reports, giving administrators and donors a real-time view of organizational performance.

6.4 Potential Risks

- Low digital literacy among some volunteers or smaller NGOs may slow adoption.
- Data-privacy concerns may arise from centralizing sensitive volunteer and beneficiary information.
- AI matching accuracy depends on the quality and completeness of volunteer profile data.
- Unreliable internet connectivity in some field locations may affect real-time synchronization.

6.5 Implementation Challenges

- Migrating existing volunteer and donation records from paper/spreadsheets into the new system.
- Training NGO staff and volunteers to adopt the new digital workflow.
- Integrating third-party services such as GIS/maps APIs and payment gateways securely.

- Ensuring the AI matching model is trained on sufficiently representative data before full rollout.

7. Conclusion

This case study analyzed the operational challenges faced by NGOs and volunteer organizations that continue to rely on fragmented, manual systems to manage volunteer registration, task assignment, donation tracking, and project monitoring. The analysis identified key stakeholders — volunteers, NGO administrators, donors, beneficiaries, project managers, and system administrators — each affected by the absence of a unified, transparent coordination mechanism, resulting in inefficient resource utilization, mismatched volunteer assignments, delayed reporting, and reduced donor confidence.

8. References

- Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson Education.
- Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.
- Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide: The Definitive Guide to Scrum — The Rules of the Game*. Scrum.org.
- Armbrust, M., Fox, A., Griffith, R., et al. (2010). A View of Cloud Computing. *Communications of the ACM*, 53(4), 50–58.
- Longley, P. A., Goodchild, M. F., Maguire, D. J., & Rhind, D. W. (2015). *Geographic Information Science and Systems* (4th ed.). Wiley.
- Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson Education.
- United Nations Volunteers (UNV) Programme. (2022). *State of the World's Volunteerism Report*. United Nations.
- World Bank. (2021). *Digital Technologies for Social Sector Organizations: A Practitioner's Guide*. World Bank Group Publications.